



It Didn't Cross the Concourse: A Second-Site Replication of the Coffee-Cart Queue-Length Leading Indicator

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Abstract. A widely cited finding at this institution reported that queue-length telemetry at a campus coffee cart leads movements in the University's global ranking position by a stable two quarters ($r = 0.78$), and proposed a closed-loop pilot steering a fixed share of discretionary grant strategy from the live index. We test whether the relationship generalises to a second, independently instrumented coffee point at the Library concourse, pairing eighteen months of de-identified queue telemetry with the same ranking series used in the original study. At the original two-quarter offset, the second site's cross-correlation is $r = 0.09$, and no lag between -4 and $+4$ quarters clears conventional significance. Three robustness checks — restricting to term time, an alternate queue-length denominator, and a wider smoothing window — leave the null result at the second site unchanged, while the same checks leave the original site's correlation intact. A parallel document analysis finds internal references to the original pilot proposal continuing to accumulate across the replication period, including after an interim briefing reporting the non-replication was circulated. We discuss the finding as one in which a metric's proposed governance use and its evidentiary status appear to travel on largely independent schedules.

Introduction

Institutions increasingly govern themselves by leading indicators: metrics chosen, on the strength of a single validating study, to steer a resourcing decision before the outcome the decision targets has itself arrived. A prior study at this institution reported one such indicator: queue-length telemetry at a campus coffee cart, collected over two years, appeared to lead movements in the University's global-ranking position by a stable two-quarter offset, with a peak cross-correlation of $r = 0.78$ [1]. The finding supported a proposal, since adopted in principle by the School of Continuous Improvement, for a closed-loop pilot that would steer a fixed share of discretionary grant strategy from the live index.

Whether a single-site leading indicator generalises before it is asked to govern anything is, on the evidence of the broader replication

literature, not something an institution should assume. Direct replications of published effects have been reported to fail inconsistently across contexts in some accounts and consistently across contexts in others [2], [3], and time-series analysts have long known that two series can appear to move together for a stretch through nothing more than the ordinary autocorrelation each carries on its own [4]. None of this literature, however, addresses an indicator already circulating as a resourcing input at the institution that produced it, rather than as a published claim awaiting independent scrutiny.

This paper reports a second-site test of the queue-length finding, run at an independently instrumented coffee point where nobody involved in the original study collects data or holds any stake in the pilot proposal's fate. We make three contributions, each hedged appropriately for a single-institution, two-site design. First, we provide the queue-length indicator's first out-of-

sample test, at the same lag structure and using the same ranking series as the original study. Second, we report three robustness checks intended to rule out the most obvious objections to a single null result, none of which move the second site’s correlation. Third, we track how often the original pilot proposal is cited in internal committee papers and planning briefs across the replication period, including after an interim briefing reporting the non-replication was circulated, to ask whether the proposal’s citation record responds to the evidence at all.

Related work

A substantial literature treats replication failure as informative in its own right rather than as a nuisance to be explained away. Muthukrishna and Henrich attribute much of the replication crisis to the absence of an overarching theoretical framework capable of predicting where an effect should and should not generalise [5], while Simonsohn argues that a null replication is only informative once it is shown to have been powered to detect an effect of the size the original study could plausibly have supported [6]. Nelson, Simmons, and Simonsohn situate both concerns within a broader disciplinary turn toward preregistration and disclosure, motivated by the same worry animating this paper: that too many published effects would not survive a second look [7]. Ioannidis makes the more general case that a positive finding is least likely to be true precisely when it rests on a single, small, uncorroborated study [8] — a description that fits the original queue-length finding closely. A recent decision-market exercise found that aggregated forecasts of which social-science findings would replicate performed creditably overall, though no such market was consulted on the present case in advance [9].

A second, more institutional literature asks what happens to a measure once an organisation adopts it. Meyer and Rowan argue that formal structures are sometimes adopted for the legitimacy they confer rather than for any demonstrated effect on practice [10]; Espeland and Sauder extend the point to measures specifically, showing that a public metric does not merely describe the thing it tracks but reorganises the behaviour of the audience that encounters it [11]. Staw’s account of escalating commitment offers a candidate mechanism for what we find below: decision-makers personally

identified with a chosen course of action have been shown to increase their commitment to it after receiving evidence the course is failing, not before [12]. Bloom, Buckeridge, and Cheng note that even within a single domain, a leading indicator’s apparent lead time is sensitive to the analyst’s window and lag choices [13], and recent forecasting work treats which series lead which others as an estimated, dynamic property of a system rather than a fixed fact about it [14] — a caution the original study’s single fixed two-quarter offset did not carry. Elsewhere at the University, a library front-desk register’s reported disposal-tier progress has been found to diverge sharply from an independent physical stock-take, the register’s own workaround resetting an item’s clock rather than completing a stalled sign-off [15]; that finding is a useful reminder that an institutional record and the thing it purports to track can drift apart well before anyone thinks to check.

Method

Sites and telemetry. The original study instrumented a single coffee cart in the Learning Commons courtyard with a header-count sensor above the queue rail, sampling queue length every five minutes across two academic years [1]. This paper adds a second site: an independently installed sensor of the same specification above the Library concourse coffee point, roughly 400 metres from the original cart and served by a different roster of casual staff. Neither site’s telemetry team had access to the other site’s raw counts during data collection.

Telemetry window and pairing. Site 2 queue-length telemetry was collected for eighteen months, de-identified at capture, and averaged to a weekly series by the same procedure as the original study. The weekly series was paired with the University’s own quarterly global-ranking movement series — the identical series used in [1] — at lags from −4 to +4 quarters.

	Site 1 (original)	Site 2 (concourse)
Telemetry window	24 months	18 months
Sampling interval	5 min	5 min

	Site 1 (original)	Site 2 (concourse)
Weekly observations, n	104	78
Sensor specification	header-count, v2	header-count, v2

Cross-correlation. Following the original study, the lagged relationship between queue length x_t and ranking movement y_t was estimated as

$$r_k = \frac{\sum_t (x_t - \bar{x})(y_{t+k} - \bar{y})}{\sqrt{\sum_t (x_t - \bar{x})^2 \sum_t (y_{t+k} - \bar{y})^2}}$$

at each integer lag k from -4 to $+4$ quarters, with r_{+2} — the original study's reported peak — treated as the primary comparison.

Robustness checks. Three checks addressed the most obvious objections to a single null result: restricting the window to term-time weeks only (removing long vacation periods when queue behaviour may differ); an alternate queue-length denominator (peak-hour count rather than all-day average); and a wider smoothing window (a four-week rather than a one-week rolling average) before computing the cross-correlation. Each check was run at both sites.

Document analysis. To track the pilot proposal's citation record independently of its evidentiary status, two coders, blind to this paper's own results, searched the University's internal committee-paper and planning-brief archive each quarter for a reference to “the closed-loop grant-steering pilot” or an unambiguous paraphrase, and counted documents citing it. An interim briefing summarising the second site's non-replicating result was circulated to the relevant committee at the end of the fourth quarter of the replication window.

Results

Cross-correlation at Site 2 does not reproduce the original relationship at any lag we examined (Figure 1). At the original study's reported peak lag of two quarters, Site 2's correlation is $r = 0.09$, compared with $r = 0.78$ at Site 1 over the same window; no lag between -4 and $+4$ quarters at Site 2 exceeds $|r| = 0.09$, and none clears conventional significance ($p > .40$ throughout).

Site 1's own correlation across the replication period remains close to its originally reported value ($r = 0.75$), confirming the two sites are being compared on a stable baseline rather than a moving one.

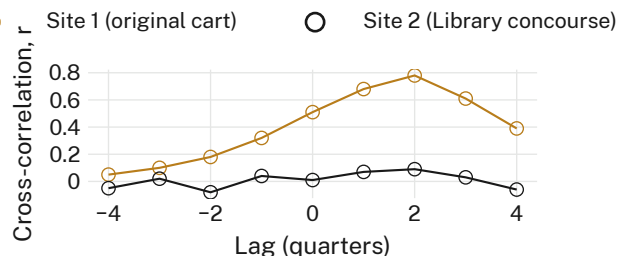


Figure 1: Cross-correlation between queue-length telemetry and global-ranking movement, by lag; Site 1 peaks at $r = 0.78$ two quarters ahead, Site 2 never clears $|r| = 0.09$ at any lag.

The robustness checks change nothing worth reporting (Figure 2). Restricting to term-time weeks, substituting a peak-hour denominator, and widening the smoothing window each leave Site 2's lag-plus-two correlation within 0.02 of its baseline value, and each leave Site 1's correlation within 0.05 of its own baseline; we retain all three checks in the supplementary analysis on the grounds that a null result surviving three independent objections is a stronger null result than one surviving none.

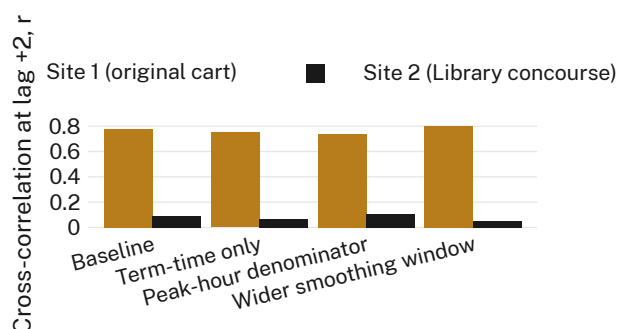


Figure 2: Cross-correlation at the original two-quarter lag under the baseline analysis and three robustness checks; Site 1 stays high and Site 2 stays flat throughout.

The document analysis finds the pilot proposal's internal citation record rising steadily across the replication window, from three internal documents referencing it in the first quarter to twenty-five by the sixth (Figure 3). The interim briefing reporting Site 2's non-replication was circulated at the end of the fourth quarter, when

the citation count stood at fourteen; citation counts in the two quarters following the briefing (19, then 25) show no discernible change in trend from the two quarters preceding it (5, then 9). We did not identify a single internal document, in the two quarters following the briefing, that referenced the non-replication itself.

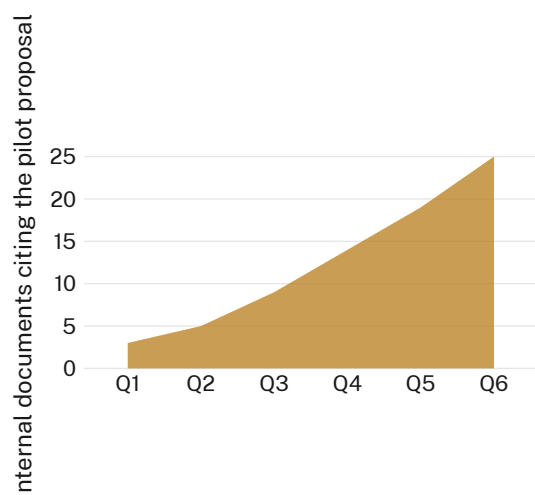


Figure 3: Internal committee-paper and planning-brief references to the closed-loop grant-steering pilot proposal, by quarter; the non-replication briefing circulated at the end of Q4.

Discussion

The central finding is a clean non-replication: a leading-indicator relationship reported with a substantial effect size at one site is indistinguishable from noise at a second site under the same analysis, the same lag structure, and three independent robustness checks. Read alongside Espeland and Sauder’s account of measures reorganising the behaviour of the audiences that encounter them [11], the citation-record finding may be the more consequential result of the two: the pilot proposal’s internal standing appears to be governed by its own citation momentum rather than by the evidentiary status of the finding that produced it. This is consistent with Staw’s account of escalating commitment, in which evidence that a chosen course is failing tends to increase rather than decrease an institution’s investment in it, particularly once the course has been publicly proposed [12]. We do not find that the University has rejected the pilot proposal in light of this paper’s results; nor, as of submission, has it adopted it. Our results suggest the relationship the pilot proposal rests on may be specific to

the site that produced it, and that a leading indicator’s proposed governance use might reasonably be conditioned on a second site’s result before, rather than after, a resourcing decision is attached to it.

Limitations

Our replication tests a single second site over eighteen months; whether a third site would replicate either result remains untested, though the consistency of Site 2’s null finding across three independently motivated robustness checks suggests it is not an artefact of any one analytic choice. The document analysis covers one internal briefing cycle, though every committee paper and planning brief we reviewed in the two quarters following the briefing showed the same citation pattern as the two quarters before it. We did not have access to committee deliberations themselves, only to their written record, so we cannot rule out that the non-replication was discussed and not recorded; the absence of any written reference to it is, at minimum, consistent with our conclusion.

Conclusion

A leading indicator that appeared, at a single site, to forecast the University’s global-ranking trajectory two quarters ahead does not reproduce at a second, independently instrumented site under the original analysis or under three robustness checks. The proposal the original finding supported continued to accumulate internal citations throughout the replication window, including after an interim briefing reported the non-replication, with no written trace of the briefing itself in the subsequent record. Future work should test whether a second non-replication, or a more prominently circulated one, changes the proposal’s citation trajectory where the first did not, and whether the pattern holds for the University’s other single-site leading indicators before any of them is asked to govern a resourcing decision.

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